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Acacia mangium

Acacia mangium is a species of flowering tree in the pea family, Fabaceae, that is native to northeastern Queensland in Australia, the Western Province of Papua New Guinea, Papua, and the eastern Maluku Islands.[3] Common names include **black wattle**, **hickory wattle**, **mangium**, and **forest mangrove**. Its uses include environmental management and wood.[2]

It was first described in 1806 by Carl Ludwig Willdenow, who described it as living in the Moluccas.[4][5]

Black wattle
Conservation status
Extinct: EX, EW; Threatened: CR, EN, VU; Least Concern: NT, LC
Least Concern (IUCN 3.1)[1]
Scientific classification
Kingdom: Plantae
Clade: Tracheophytes
Clade: Angiosperms
Clade: Eudicots
Clade: Rosids
Order: Fabales
Family: Fabaceae
Subfamily: Caesalpinioideae
Clade: Mimosoid clade
Genus: *Acacia*
Species: ***A. mangium***
Binomial name
Acacia mangium

Cultivation

Acacia mangium grows up to 30 metres (98 ft), often with a straight trunk. *A. mangium* has about 142,000 seeds/kg.[6] To break down dormancy mature seed requires pre-germination treatments, such as mechanical scarification (scratching the surface) or boiling water. This treatment leads to a fast germination and typically exceeds 75%.[7] Like many other legumes, it is able to fix nitrogen in the soil.[8] *A. mangium* is a popular species for forest plantation and agroforestry projects. In mixed cultures, plants can profit from shading by *A. mangium* and its nitrogen fixation.[9] *A. mangium* will tolerate low fertility soils with impeded drainage, but prefers fertile sites with good drainage. Soil depth and topographic position can influence yields. With respect to distance from the equator, there are significant differences in performance under cultivation. A mean annual height increase of about 3 to 4 m is usual near the equator, whereas in areas further from the equator growth is slower.[7]

Uses

Timber

Acacia mangium trees produce sapwood and heartwood. The heartwood's colour is brownish yellow shimmery and medium textured. Because the timber is extremely heavy, hard, very strong, tough, and not liable to warp and crack badly, it is used

for furniture, doors and window frames. The glossy and smooth surface finish after polishing leads also to a potential for making export-oriented parquet flooring tiles and artifacts.[10]

Pulp and paper

A. mangium has been recognized as an excellent source of short cellulose fibers for papermaking. It is grown in plantations in Southeast Asia, particularly Indonesia.[11]

Soil management and ecological restoration

Because *Acacia mangium* trees increase the turnover rate of nitrogen in the topsoil, it can improve the nitrogen availability in soils in mixed cultures.[12] Due to the fact that it is a very fast growing tree, it develops an intensive rooting system, particularly in low fertility soils.[13] This helps to recover degraded tropical lands.[12] The tree is widely used in Goa, India, in the mining industry for rehabilitation of waste dumps, as it is a drought-resistant species and binds sterile mine waste consisting of lateritic strata.[6] In Colombia, it has been used for restoring wasteland created by open-pit gold mining.[14]

Willd.

Occurrence data from AVH

Synonyms

- *Acacia glaucescens sensu* Kaneh. & Hatus.
- *Mangium montanum* Rumph.
- *Racosperma mangium* (Willd.) Pedley[2]

Chemistry

The gum contains 5.4% ash, 0.98% N, 1.49% methoxyl, and by calculation, 32.2% uronic acid.[15] The sugar composition after hydrolysis: 9.0% 4-0-methylglucuronic acid, 23.2% glucuronic acid, 56% galactose, 10% arabinose, and 2% rhamnose.

Gallery

Cultivation in Africa

Trunk in Kolkata, India

Tree in Kolkata, India.

Leaves with fruit pod at canopy

Leaves and pods

Old bark

Leaves in young tree

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